

M.E. Sharpe

A Post Keynesian Theory of Credit Rationing

Author(s): Martin H. Wolfson

Reviewed work(s):

Source: *Journal of Post Keynesian Economics*, Vol. 18, No. 3 (Spring, 1996), pp. 443-470

Published by: [M.E. Sharpe, Inc.](#)

Stable URL: <http://www.jstor.org/stable/4538500>

Accessed: 24/09/2012 01:41

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



M.E. Sharpe, Inc. is collaborating with JSTOR to digitize, preserve and extend access to *Journal of Post Keynesian Economics*.

<http://www.jstor.org>

A Post Keynesian theory of credit rationing

It has become clear in the 1980s and 1990s that interruptions in the supply of credit can have important macroeconomic implications. The Federal Reserve intervened in the wake of financial crises (such as the 1987 stock market crash and the 1991 failure of the Bank of New England) to insure that the flow of credit was maintained. The “credit crunch” of 1990–92 brought to national attention once again the importance of credit rationing and the central role of access to credit for economic expansion.

In the wake of these developments, the economics profession has scrambled to keep up. The emergence of the “credit view” in macroeconomic theory has argued for the importance of credit, and for financial variables more generally, in macroeconomic outcomes. The unavoidable importance of the financial system in interpreting recent economic history has encouraged the acceptance of the credit view. It has gained stature compared with neoclassical models in which the financial system enters only in the form of the supply of money, and especially compared with abstract Walrasian general equilibrium models. As is well known, the only appearance for anything remotely financial in Walrasian models is money’s role as a numeraire (and even then any other commodity can function in that capacity as well).

The greater realism afforded by the credit view has been a welcome addition to macroeconomic theory. However, much of the edifice of the credit view has been built upon a specific assumption, that of asymmetric information. New Keynesian economists have applied this assumption widely to include a number of situations (such as the labor market and stockholder–manager interactions) in which the presence of princi-

The author is in the Department of Economics, University of Notre Dame, Notre Dame, Indiana. For helpful comments on an earlier draft of this paper, he would like to thank Robert Blecker, Paul Burkett, Jim Crotty, Amitava Dutt, Gary Dymski, Ilene Grabel, Peter Skott, and Ed Slattery. Any errors are the responsibility of the author.

pal-agent problems suggests that different parties may have access to different—that is, asymmetric—information.

Nonetheless, the asymmetric information assumption remains relatively narrow. It ignores most of the Post Keynesian literature on uncertainty, time, and expectations formation. So the objective of this paper is to investigate the phenomenon of credit rationing from this broader Post Keynesian perspective. After a brief discussion of the asymmetric information approach in the next section, the third section discusses those elements of Post Keynesian theory relevant to credit rationing. The fourth section examines some institutional features of bank lending. Finally, the fifth section attempts to build a somewhat more formal conception of credit rationing that is faithful to both institutional realities and Post Keynesian theory.

Asymmetric information and credit rationing

The New Keynesian theory of credit rationing assumes that there is a precise probability distribution of returns from each project that a potential bank borrower wants to undertake. This distribution is known by the borrower, but not by the bank. Although it may know the expected mean return of a project, “the bank cannot ascertain the riskiness of a project” (Stiglitz and Weiss, 1981, p. 395).

Nonetheless, the bank assumes that increasing the interest rate it charges borrowers may adversely affect the riskiness of the borrowers’ projects. The riskiness may change for two reasons, the adverse selection effect and the incentive (or moral hazard) effect. The adverse selection effect refers to the idea that those borrowers willing to pay a higher interest rate may be worse risks. Borrowers with riskier projects perceive their probability of repayment to be low, and so are willing to pay a higher interest rate. In other words, their projects are unlikely to pay off, but if they do, they will pay off with large enough returns to easily cover the interest costs of the loan. On the other hand, the relatively safe borrowers are assumed to drop out of the market. Since their distribution of returns is relatively narrow, even their best possible outcomes are too small to be able to repay the higher interest rate.

The incentive effect refers to the bank’s assumption that higher interest rates will influence a borrower’s choice of projects. As interest rates increase, any given borrower will have a tendency to choose riskier projects: those that have a higher payoff when successful (and a lower payoff when unsuccessful). The rationale is that they will seek out those

projects with potential returns high enough to repay the higher interest charges.

One might think that the higher-risk projects would deter borrowers who fear bankruptcy. However, in these models it is typically assumed that borrowers are risk-neutral (they consider only their expected return and not their risk), the lower payoff always results in default, projects are considered in isolation from the overall fortunes of the borrower, and there are no bankruptcy costs. Under these conditions, choosing a riskier project will increase the expected return to the borrower of a successful outcome, but will not decrease the return if the outcome is unsuccessful (it will remain at zero). Since the project is considered in isolation from the borrower's overall financial condition, a default on the loan is equivalent to bankruptcy; it is not possible for a larger loss to push the borrower closer to bankruptcy. Without these assumptions, a borrower might not necessarily choose the riskier project. (Similar considerations apply to the adverse selection effect.) So the empirical relevance of the theory might be relatively narrow.

Nonetheless, given the assumptions, it will be the case that increasing the interest rate will attract borrowers of higher risk, or induce borrowers to choose projects with higher risk. Both of these developments are assumed to reduce the bank's expected profits because the bank, receiving a contractually fixed return, does not benefit from higher borrower profits, but does lose if profits fail to provide for interest payments. Of course, there is a direct positive effect on bank profits of higher interest rates, but Jaffee and Stiglitz (1990, p. 857) argue that

the adverse selection effect [and the moral hazard effect] can easily outweigh the direct effect. . . . There is a critical interest rate, r^* , at which the safer borrowers stop applying. Thus, the return to the bank declines precipitously at r^* .

The bank might choose to charge a higher interest rate to these high-risk borrowers, but because of asymmetric information, is unable to distinguish the "good" (low-risk) borrowers from the "bad" (high-risk) borrowers.¹ The solution the bank chooses to maximize its profits under these conditions is to ration credit. The type of credit rationing

¹ As Crotty (1994) has pointed out, despite its inability to distinguish between high-risk and low-risk borrowers, the bank has to know an enormous amount about its borrowers in order to calculate r^* , the rate at which the safe borrowers drop out of the market. The bank needs to know precisely how these borrowers will react to increases in the interest rate.

that arises under conditions of imperfect information is termed “pure” credit rationing in Jaffee and Stiglitz’s review of the credit rationing literature. They define it as “instances in which some individuals obtain loans, while *apparently identical individuals*, who are willing to borrow at precisely the same terms, do not (Jaffee and Stiglitz, 1990, p. 849).²

As Stiglitz and Weiss note, “When the price (interest rate) affects the nature of the transaction, it may not also clear the market” (Stiglitz and Weiss, 1981, p. 393). Although the interest rate chosen by the bank to maximize profits is an equilibrium rate, it is nonetheless the case that the supply of credit does not equal the demand for credit.³

This situation is illustrated in figure 1. The bank’s supply curve is backward-bending. At relatively low interest rates such as r_1 , supply will equal demand at L_1 , and there will be no credit rationing. If the supply of loanable funds to the bank decreases (because of a tighter monetary policy or reduced deposit inflows), the supply curve will shift leftward, and demand and supply will be equal at an interest rate of r_2 .⁴ However, the bank will refuse to raise its interest rate beyond r^* . Because of adverse selection and incentive effects, the bank’s expected return is maximized at r^* . It will therefore ration credit, despite the willingness of borrowers to pay a higher interest rate. The equilibrium interest rate is r^* and there is an excess demand for loans given by L_2L_3 .⁵

The credit view

Other New Keynesians, working primarily in the macroeconomic context, have used the same assumption of asymmetric information to arrive at a somewhat different result. These theories assume that asymmetric information causes problems for all lenders. But instead of focusing on

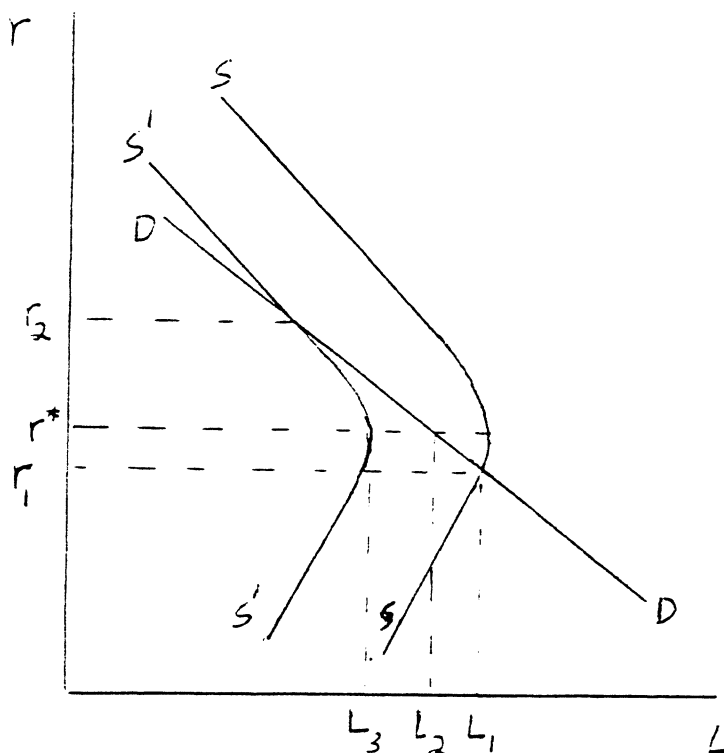
² Since by assumption the bank cannot distinguish the high-risk borrowers from the low-risk borrowers, they are of necessity apparently identical. The authors do not explain, however, how the bank decides who among these apparently identical borrowers receive credit and who do not.

³ As Stiglitz (1987) has shown, the condition that the quality of a product is influenced by its price is capable of “repealing” the law of supply and demand in many different contexts.

⁴ As drawn, r^* , the interest rate that maximizes the bank’s expected return, remains unchanged under the new supply conditions. This will be the case as long as the probability distributions describing the borrowers’ projects remain unchanged. See Stiglitz and Weiss (1981, p. 255) and Jaffee and Stiglitz (1990, p. 861).

⁵ See Stiglitz and Weiss (1981) for a derivation of the supply curve for bank loans from the relationship between loans extended and the mean return (to the bank) on bank loans.

Figure 1 Credit rationing with asymmetric information



the bank's problems as representative of the problems of all lenders, these theories emphasize the *relative* ability of banks to overcome the incomplete information, compared with other lenders. Thus, banks are "special" in the sense that they have expertise in gathering information about borrowers, evaluating projects, and monitoring borrowers after receipt of a loan.⁶

The significance of this expertise is that some borrowers may be able to borrow at banks, whereas they are either unable to borrow from other lenders or can borrow only at increased cost. This implies that anything that disrupts normal banking activity can reduce the supply of credit to those borrowers dependent on banks, and thus have macroeconomic implications. This insight has spawned the burgeoning "credit view"

⁶ Here "banks" are taken to be all those intermediaries (including pension funds and brokerage houses) that have a specialized information-gathering ability. "Banks" of this sort are contrasted with lenders in "arm's-length" markets such as the commercial paper market.

literature, which has been applied to explain declines in economic activity due to restrictions in bank lending in a number of contexts. These include bank runs during the Great Depression, credit controls, capital constraints, and tight monetary policy. In particular, its application to monetary policy has opened up a “credit channel” that works through a decline in the supply of bank loans rather than (or in addition to) a fall in the money supply or the indirect effect on aggregate demand of higher open market interest rates (brought about through a lower money supply).

It is argued that the existence of a credit channel is not dependent on credit rationing. Bernanke explains:

Credit rationing . . . is certainly *consistent* with the existence of a credit channel. . . . However, credit rationing is not at all *necessary* for the credit channel to exist. All that is required for a credit channel is that bank credit and other forms of credit be imperfect substitutes for borrowers. [Bernanke, 1993, p. 56]

In fact, in discussing the early 1990s’ “credit crunch,” which many have assumed involved credit rationing, Bernanke and Lown (1991, p. 207) say: “We define a bank credit crunch as a significant leftward shift in the supply curve for bank loans, holding constant both the safe real interest rate and the quality of potential borrowers.”

Of course, if bank-dependent borrowers are unable to borrow at all from other lenders, then presumably they would be credit-rationed by those lenders (although not by the bank), and presumably the reason would be asymmetric information. But since their overall spending would be reduced if they are unable to borrow as cheaply from non-banks as from banks, then credit rationing is not required, as Bernanke contends.

A Post Keynesian perspective

How would one go about analyzing the phenomenon of credit rationing from a Post Keynesian perspective? The following ideas would play an important role.

The fringe of unsatisfied borrowers

One of the earliest references to credit rationing in Keynes’s writings is to be found in the *Treatise* (Keynes, 1930, vol. I, p. 212):

There is apt to be an unsatisfied fringe of borrowers, the size of which can be expanded or contracted, so that banks can influence the volume

of investment by expanding or contracting the volume of their loans, without there being necessarily any change in the level of bank-rate, in the demand-schedule of borrowers, or in the volume of lending otherwise than through the banks. This phenomenon is capable, when it exists, of having great practical importance.

The implication here is that the unsatisfied fringe represents a permanent situation: there is always a group of borrowers who are refused credit. Keynes also implies that an important way that a bank determines which of its potential borrowers will receive credit and which will become a part of the "unsatisfied fringe," has to do with the relationship the borrower has with the bank: "the amount lent to any individual [is] governed not solely by the security and the rate of interest offered, but also by reference to the borrower's purposes and his standing with the bank as a valuable or influential client" (Keynes, 1930, vol. II, p. 365). Thus, presumably, those long-standing or otherwise valuable customers of a bank will be less likely to be rationed.⁷

Borrower's risk and lender's risk

An important concept is Keynes's notion of borrower's and lender's risk. In the *General Theory* (Keynes, 1936, p. 144), he defines them as follows:

Two types of risk affect the volume of investment . . . The first is the entrepreneur's or borrower's risk and arises out of doubts in his own mind as to the probability of his actually earning the prospective yield for which he hopes. . . .

But where a system of borrowing and lending exists, by which I mean the granting of loans with a margin of real or personal security, a second type of risk is relevant which we may call the lender's risk. This may be due either to moral hazard, i.e., voluntary default or other means of escape, possibly lawful, from the fulfillment of the obligation, or to the

⁷ This insight seems to be at variance with the asymmetric information approach: wouldn't the borrower's relationship with the bank be one way of distinguishing between good and bad borrowers, which the bank is supposed to be incapable of doing? Actually, New Keynesian writers contend that the bank is capable of distinguishing among many groups within the population; "pure" credit rationing then occurs only *within* groups. Although in the literature one can find references to "risk categories," the typical presentation assumes that the bank is able to sort borrowers according to the mean return of their projects (which of course does not correspond to the same degree of risk). Nonetheless, the basis for categorizing these groups has generally remained fairly vague.

possible insufficiency of the margin of security, i.e., involuntary default due to the disappointment of expectation.

The aspect of lender's risk that refers to moral hazard clearly supports an asymmetric information perspective. Indeed, moral hazard is commonly mentioned in the New Keynesian literature as one implication of asymmetric information. It is reasonable to assume that the borrower has more information than the lender concerning the possibility of voluntary default.

Nonetheless, there is no obvious reason why borrower's risk and that part of lender's risk that is not concerned with moral hazard are due to asymmetric information. These risks have to do with the disappointment of expectations, the inability to predict the future. The most appropriate Post Keynesian assumption is that both borrower and lender are subject to fundamental uncertainty about the future.

An uncertain future

By fundamental uncertainty, of course, is meant the idea that the future cannot be reduced to an actuarial calculation of probabilities, whether objective or subjective. About many important future events, we simply "do not know." This idea has been discussed and analyzed in depth by many Post Keynesian economists (Davidson, 1991; Crotty, 1994), so there is no need to elaborate on it further here. The main significance for our discussion, though, is the following: although the borrower may know some things that the lender does not know (and indeed the lender may know some things that the borrower does not know), both of them are subject to fundamental uncertainty about the future.

Asymmetric expectations

Although both borrower and lender are subject to fundamental uncertainty, it is not necessarily the case that both will come to the same conclusion about the future profitability of any particular project. Indeed, the most likely outcome is that they will come to different conclusions.

It may be thought that the lender will be more risk-averse than the borrower; it is, of course, the lender's funds that are being lent. James Tobin, in fact, has argued that this difference in attitude toward risk is typical of borrowers and lenders and that it forms the basis for credit rationing:

Typically [borrowers'] indebtedness is rationed by lenders, not just because of market imperfection but because the borrower has greater

optimism about his own prospects and the value of his collateral, or greater willingness to assume risk and to die insolvent, than the lender regards as objectively and prudently justified. [Tobin, 1980, p. 10]

Such a difference in risk preference between borrower and lender may indeed be an accurate characterization. However, all that is necessary for a theory of credit rationing based on Keynesian uncertainty is that borrower and lender evaluate the future differently—that is, that they have *asymmetric expectations*.

In fact, assume that there is absolutely no difference in risk preference between borrower and lender. Given any event with a defined risk, both will evaluate it in exactly the same way. However, because the future is uncertain, borrowers and lenders do not necessarily agree on the riskiness of any particular project whose outcome depends on the future.

Let us also assume that both borrowers and lenders have the same information about the future. To keep the example simple, we will assume that there is complete ignorance about the future. Thus, borrowers and lenders have the same risk preference and the same information (ignorance), but they evaluate that information differently (they have asymmetric expectations). In addition, we will make the clearly unrealistic assumption that borrowers and lenders will agree on the riskiness of any given project only by chance; both will essentially flip a coin to decide whether any particular project is safe enough to pursue. (We are also implicitly assuming that both borrowers and lenders are risk-averse; given the return to a project, there is a finite limit to the risk willingly incurred by either borrower or lender.)

Thus, given identical attitudes toward risk, there is a fifty-fifty chance that the borrower will decide it is too risky to pursue, and a fifty-fifty chance that the lender will also reject it. Therefore, if there are n possible projects, one-fourth (or $n/4$) will be rejected by both borrower and lender, one-fourth will be accepted by both, one-fourth will be accepted by the borrower but rejected by the lender, and one-fourth will be rejected by the borrower but not by the lender.

One-half of the projects will be judged by the borrower as having an unacceptable borrower's risk. Although half of these would have been funded by the lender, the borrower will not attempt to fund them. Of the projects deemed safe enough by the borrower, half will be rejected by the lender as involving an unacceptably high lender's risk. The results are summarized in Table 1.

Only those projects deemed safe by both borrower and lender (those

Table 1
The effect of asymmetric expectations

		Borrower	
		Safe	Risky
Lender	Safe	Lend	****
	Risky	Ration	****

in the top left cell in the table) will be funded. For those projects deemed safe by the borrower but too risky by the lender (those in the bottom left cell), the borrower will be rationed. Those projects in the right-hand column (perceived by the borrower to involve an unacceptably high borrower's risk) *will never be seen by the lender*.

Thus, to an outside observer (or to the lender), it may appear that the borrower has a greater tolerance for risk than the lender has. However, by assumption, their risk preferences are equal. It is only their evaluations of an uncertain future that differ. Under more realistic assumptions, of course, both borrower and lender evaluate the future with methods other than chance, and their risk preferences are likely to differ. However, the key point remains: their expectations of an uncertain future will differ (will be asymmetric), and therefore some projects deemed safe by the borrower will be judged as too risky by the lender. An "unsatisfied fringe of borrowers" will be subject to credit rationing.

The qualitative results of Table 1 will remain in force. In the more realistic case, the cells in the table would not be composed of equal numbers of projects. But the analysis would not in other respects be different.

Uncertainty and the lending decision

What can we say, though, about the more realistic case? How does uncertainty affect the criteria that bankers use in forming judgments about the risk of repayment? Here again we can look to Keynes's analysis of decision making under uncertainty:

In practice we have tacitly agreed, as a rule, to fall back on what is, in truth, a *convention*. The essence of this convention—though it does not, of course, work out quite so simply—lies in assuming that the existing state of affairs will continue indefinitely, except in so far as we have specific reasons to expect a change. This does not mean that we really believe that the existing state of affairs will continue indefinitely. We

know from extensive experience that this is most unlikely. . . . We are assuming, in effect, that the existing market valuation, however arrived at, is uniquely *correct* . . . though, philosophically speaking, it cannot be uniquely correct, since our existing knowledge does not provide a sufficient basis for a calculated mathematical expectation. [Keynes, 1936, p. 152]

In other words, we know that we do not know the future. But we act as if we do, by relying on conventions. Keynes also points out that these conventions can be held with varying degrees of confidence:

The state of long-term expectation, upon which our decisions are based, does not solely depend, therefore, on the most probable forecast we can make. It also depends on the *confidence* with which we make this forecast—on how highly we rate the likelihood of our best forecast turning out quite wrong. [1936, p. 148]

Keynes termed this the *state of confidence*.

There are several implications of Keynes's views that could be applied to an analysis of the bank lending decision. First, despite the existence of uncertainty, bankers do form opinions about the likelihood of repayment.⁸ These opinions are influenced by various conventions, and they may be firmly or weakly held, but they are opinions nonetheless. And they form the basis for the lending decision.

Second, the lending convention depends on an initial assumption that the existing state of affairs will continue into the future. Thus, borrowers who have a history of repaying loans on time and maintaining a strong financial condition will be preferred.

Third, lending conventions can be held with varying degrees of confidence. This is especially relevant to conditions in the macrofinancial environment. As these conditions change, a bank may revise its opinion about the riskiness of a potential loan—that is, the assumption that the future will look like the past will be maintained only until there is a reason to think otherwise. If the state of confidence is weak, the opinion is more likely to change.

This observation raises two further questions. First, how major and how widespread will the change in bankers' opinions be in response to the change in conditions? Second, do these opinions change in any systematic way?

⁸ As noted above, New Keynesian theories, despite their emphasis on the inability of bankers to judge the risk of projects *within* a defined group, also claim that bankers can sort borrowers into different groups based on risk.

Regarding the change in bankers' opinions, Keynes's analysis of the stock market also could be applied to the banking context:

A conventional valuation which is established as the outcome of the mass psychology of a large number of ignorant individuals is liable to change violently as the result of a sudden fluctuation of opinion due to factors which do not really make much difference to the prospective yield; since there will be no strong roots of conviction to hold it steady. [Keynes, 1936, p. 154]

These changes in opinion could be expected to affect most bankers simultaneously. It is in the nature of a convention that most people adhere to it and few wish to be left out.

Financial fragility

Relevant to systematic changes in bankers' opinions is the endogenous development of financial fragility. A perceived change in the financial condition of bank borrowers will be likely to change bankers' conventional valuations of the risk of lending, and thus the extent of credit rationing. As Hyman Minsky (1986, p. 213) has argued, this change takes place endogenously:

[I]n a world of uncertainty, given capital assets with a long gestation period, private ownership, and the sophisticated financial practices of Wall Street, the successful functioning of an economy within an initially robust financial structure will lead to a structure that becomes more fragile as time elapses. Endogenous forces make a situation dominated by hedge finance unstable, and endogenous disequilibrating forces will become greater as the weight of speculative and Ponzi finance increases.

This endogenous increase in financial fragility is especially likely to take place over the course of a business-cycle expansion and over a series of relatively mild business cycles.

Financial fragility increases as borrowers take on more debt, as the maturity of that debt shortens, and as liquidity declines. Minsky (1977a) argues that rising interest rates can push a financially fragile system toward financial crisis by increasing cash payment commitments relative to cash receipts and by reducing the market value of assets relative to liabilities (assuming that assets are of longer term than maturities). Both of these developments will lead to changes (and often sudden changes) in bankers' attitudes toward the riskiness of lending, to what Crotty (1994) calls the *endogenous development of expectations*, and to a restriction in lending.

Note that the increase in interest rates, by making debt repayment more difficult and by eroding the value of collateral, can do exactly what Stiglitz (1987) refers to in the asymmetric information context: make the quality of a product (the promise to repay debt) dependent on its price (the interest rate). The result in this case is the same: increased credit rationing.

If bankers' attitudes change endogenously so that they become less willing to lend, are borrowers' attitudes also changing so that they become less willing to borrow? In that case, we might not expect to see an increase in credit rationing. Although borrowers may at this point be less willing to finance new investment projects, it is not necessarily the case that they have a reduced demand for credit. Increased financial fragility is likely to lead to an increased need to borrow: "speculative and especially Ponzi finance give rise to large increases in an interest-inelastic demand for finance" (Minsky, 1980, p. 221). In addition, the practice of sequential financing of investment (raising funds as the project proceeds) contributes to the inelastic demand for finance (Minsky, 1977b, p. 149). This "necessitous borrowing" has played an important role near the business-cycle peaks in postwar U.S. history (Wolfson, 1994, pp. 167–171).

Thus, there is likely to be a consistent increase in credit rationing as the peak of the business-cycle expansion approaches; the degree of credit rationing depends at least in part upon the stage of the business cycle (and of course the increase in credit rationing also contributes to the ending of the business-cycle expansion and the beginning of the recession). So static models of credit rationing that do not take account of history, of the *irreversible past*, are unlikely to describe reality accurately.

Endogenous money

Before we leave this discussion of Post Keynesian theory, it may be useful to address the issue of endogenous money. At first glance, the two concepts of endogenous money and credit rationing seem incompatible. If bankers create money by accommodating the credit demands of bank borrowers, how can bankers also refuse to accommodate these demands by rationing credit?

The answer is simply that bankers accommodate all *creditworthy* demands for credit, and ration all those demands not deemed creditworthy. The implications of this perspective will be explored further in the last section.

Institutional aspects of bank lending

Before we attempt to develop these ideas more precisely, it might be useful to examine some institutional aspects of bank lending relevant to credit rationing. The Federal Reserve's Survey on Bank Lending Practices and Survey on the Terms of Bank Lending are important sources of information in this regard. We will use these surveys to shed light on several important questions, in particular, a bank's ability to judge the relative risk of borrowers, how risk and other factors enter into the prices charged borrowers, the importance of an ongoing relationship between borrower and bank, how banks ration credit, and the relationship between credit rationing and interest rates.

Do banks judge the risk of borrowers?

The Federal Reserve's Senior Loan Officer Opinion Survey on Bank Lending Practices asks banks a series of questions about their lending practices. In the November 1994 survey, fifty-seven domestic commercial banks and twenty-four U.S. branches and agencies of foreign banks were asked about the extent to which they evaluated the riskiness of their business borrowers. The specific question was, "In the process of reviewing applications, does your bank 'risk-grade' firms applying for commercial and industrial loans—i.e., does it assign applicants to one of a number of possible risk categories, rather than simply 'accept' or 'reject'?" (Board of Governors, 1994, p. 13). The responses indicated that judgments about the riskiness of borrowers were nearly always made: "Virtually all of the respondents indicated that they categorize borrowers of all sizes based on risk" (Board of Governors, 1994, p. 1). About one-fifth of the respondents used a statistically based credit scoring model to assign borrowers to risk categories. For the most part, risk evaluation was done on a judgmental basis.

How banks price loans

In addition, the report noted that "A large majority of the respondents indicated that price and non-price terms are varied based on a borrower's risk" (Board of Governors, 1994, p. 4). Here "price" terms refer to explicit charges such as the cost of credit lines or the spread of loan rates over a base rate; non-price terms are other aspects of the loan agreement such as requirements for collateral, loan covenants, or the size of credit

lines.⁹ So a borrower that the bank judges to be relatively risky would be charged a higher interest rate than a less risky borrower, and would also be subject to higher noninterest terms.

In determining the interest rate on business loans, banks use two different methods. The traditional method uses an administered rate, the prime rate, as the reference rate. Borrowers then pay an interest rate that is a specific spread over the prime rate. Interest rates on most small loans are determined in this way.

Most large business loans, however, use money market rates as the reference rate. These loans, which make up most of the dollar volume of all business loans made by banks, are generally made at interest rates below prime. They are determined in two different ways: credit lines and loan commitments. As explained by Brady (1985, pp. 1–2):

Lines are relatively informal arrangements under which banks agree that on demand they will quote a price on a fixed-rate loan for a particular amount and maturity (usually under a year) selected by the borrower, within specified limits. Banks typically set the price of such loans by marking up a reference market rate for the same maturity as the loan—for example, the rate on 30-day certificates of deposit for a 30-day loan . . . By widening or narrowing their markups, banks can affect the quantity of these loans outstanding.

Unlike lines, loan commitments require fees, which are paid on the unused credit available under the arrangement. Commitments oblige banks to lend up to agreed-upon amounts as long as borrowers meet loan covenants included in the contract. . . . borrowers draw down credit up to specified limits at preestablished spreads over reference rates.

Thus, there are three types of lending procedure, each with a different degree of relative control by the borrower and the bank over the interest rate charged and the amount borrowed. Under prime rate pricing, the bank retains the most control. It sets not only the spread over the prime, but also the prime rate itself. It retains the most flexibility with regard to the amount of the loan. With credit lines, the borrower is able to determine, within specified limits, the amount to be borrowed, but the interest rate (the spread over market rates) is determined by the bank. However, by varying the spread the bank can influence the amount of credit extended. Finally, with loan commitments the borrower determines both the amount borrowed (subject to the size of the unused commitment) and the interest rate paid (subject to preestablished agreement).

⁹ See below for a discussion of credit lines.

The importance of an ongoing relationship

For many years, the Federal Reserve's Survey of Bank Lending asked banks about their "credit standards" for loans—that is, their price and nonprice terms. From the above discussion, it is reasonable to conclude that these standards were affected by changing perceptions of risk, and that they affected different borrowers in different ways; the brunt of the tightening or easing of standards fell mostly on borrowers without loan commitments.

We would assume that borrowers with loan commitments are the bank's best customers, the ones with whom the bank has maintained an ongoing successful relationship. As noted earlier, one of the implications of Keynes's views on uncertainty is that borrowers who have maintained a successful borrowing record with the bank in the past will be preferred. Thus, we would expect that, at any given time, borrowers with an established borrowing record would be subject to less tightening of credit standards than would borrowers without such a record.

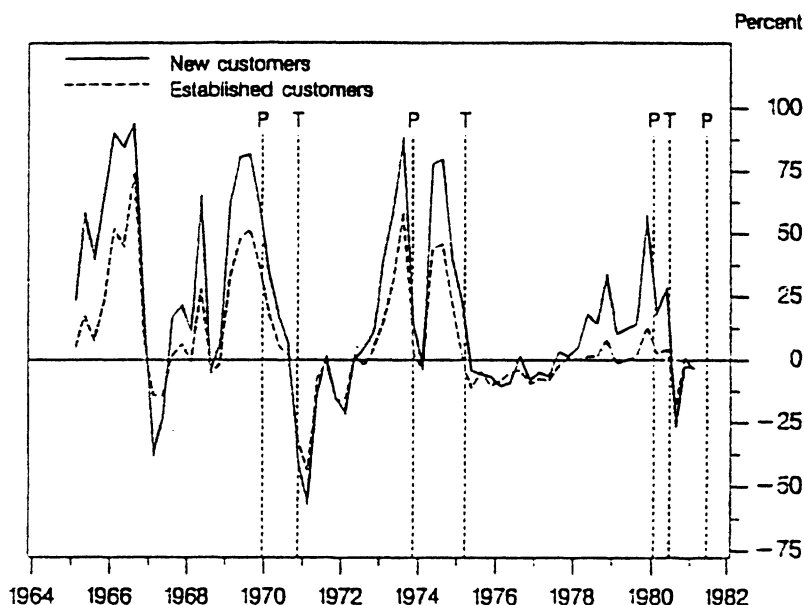
Figure 2 indicates that this is indeed the case. It plots the results of a question asked continuously on the Survey of Bank Lending Practices from 1965 through 1981. (Unfortunately the question in this form was discontinued in 1982.) The question asked a sample of banks (totaling 122 for most of those years) how they had changed their credit standards compared with those three months earlier. The answers were recorded for two classes of borrowers, new and established (those with a history of borrowing from the bank). The data in the figure indicate the net percentage of banks that had tightened these standards, that is, the number of banks that tightened minus the number that loosened, as a percentage of all banks surveyed. It is clear from the figure that the standards for established borrowers were almost always subject to less tightening than those for new borrowers.

How banks ration credit

Although new borrowers were subject to a greater tightening of credit standards than established borrowers, what does this say about credit rationing? Isn't a tightening of standards simply an increase in the requirements associated with existing loans, and an increase in the costs of borrowing on these loans, rather than a refusal on the part of banks to lend? So figure 2 might not necessarily be measuring the degree of credit rationing.

It is indeed true that tighter standards increase the costs and require-

Figure 2 Bank credit standards



ments of loans that the bank actually makes. However, such standards also have another function: to raise requirements necessary to achieve in order to obtain a loan. If a borrower is unable to post sufficient collateral, it is unlikely that the loan will be made. Also, typically banks have standards in the form of minimum income and cash flow requirements, maximum loan-to-value ratios, and so on. Those borrowers below the minimum (or above the maximum) will not obtain the desired loan.

In addition, one important nonprice term concerns the size of credit lines. Thus, tighter standards imply a direct form of credit rationing, not by denying the loan request outright, but by limiting the size of the loan. For both these reasons, figure 2 can be interpreted to show that banks whose risk preferences have changed are able to increase their credit rationing by raising their standards for making a loan.

So, to summarize, the evidence from the bank lending surveys discussed above leads to the following conclusions:

1. Banks classify borrowers according to perceived risk, and use these risk classifications to set both price and nonprice terms of lending.

2. Higher price terms imply a higher spread over the reference rate, except for borrowers with loan commitments (in the short run).
3. Higher nonprice terms raise requirements on existing loans, but also provide the basis for denying credit to those borrowers judged to be insufficiently creditworthy. And
4. Banks give preference to borrowers with whom they are familiar.

Credit rationing and interest rates

The above discussion leads to an important conclusion: since banks increase both price and nonprice terms in response to perceptions of higher risk, and since higher nonprice terms are the basis for denying a loan, we would expect to see spreads over reference rates and credit rationing move together. This is interesting because for those borrowers who are rationed, the bank has denied them credit rather than increase the interest rate charged.

In addition, we would generally expect banks to increase price and nonprice terms when their reserves are under pressure from tight monetary policy. Since short-term interest rates increase when monetary policy tightens, we would expect to see interest rates (in addition to the spread over a reference rate) and credit rationing also move together. It is interesting that this is a different outcome than that predicted by either of the asymmetric information models discussed earlier. In the "pure" credit rationing case, there is an increase in credit rationing without an accompanying increase in interest rates or interest-rate spreads.¹⁰ In the credit view, a leftward shift in a traditional upward-sloping bank supply curve results in an increase in interest rates, but without any credit rationing.

An examination of figure 2 indicates that banks have generally tightened their standards, and presumably also their credit rationing, as the peaks of the business-cycle expansion have approached. In addition, Lang and Nakamura (1991), examining data for 1979–91, found that the percentage of bank "safe" loans (defined as those made at interest rates below the prime rate plus one percent) was highly cyclical, and rose prior to all three business-cycle peaks during that time period. If we

¹⁰ It is the case, though, that when the basic analysis is extended to consider "*n* observationally distinguishable groups," then an increase in the cost of loanable funds can lead to an increase in credit rationing (Stiglitz and Weiss, 1981, pp. 268–269). However, the increase in the cost of loanable funds does not automatically translate into an increase in the average loan rate charged all groups, this credit rationing is not the "pure" type due to asymmetric information, and the basis for distinguishing among these groups still remains vague.

interpret the increase in the proportion of safe loans as indicative of greater credit rationing of those borrowers unable to borrow at preferred interest rates, then this evidence also supports the view that credit rationing typically increases near the business-cycle peak.¹¹ Since, in the postwar U.S. experience, monetary policy has typically tightened prior to business-cycle peaks, and since tighter monetary policy results in higher short-term interest rates, it has generally been the case that interest rates and credit rationing have increased together near business-cycle peaks.

However, the most recent business-cycle peak (July 1990) is an exception (Friedman and Kuttner, 1994; Wolfson, 1994, pp. 229–231). For a year prior to this peak, the Federal Reserve was lowering, not increasing, the federal funds rate, and interest rates continued to fall until the beginning of 1994. However, during 1990 and 1991 (data are not available for 1989) banks were clearly tightening their credit standards (Hamdani, Rodrigues, and Varvatsoulis, 1994; Wolfson, 1994, pp. 178–179), and in late 1990 and early 1991 banks significantly increased their lending spreads (Board of Governors, 1994, p. 8).

Also, in 1994, as the Federal Reserve began to push up the federal funds rate again, there were widespread reports of decreased credit rationing and lower spreads (Hansell, 1995). Therefore, it seems that interest-rate spreads, nonprice standards, and credit rationing increase together; the general level of interest rates often increases also, but not always. So the challenge of formulating a Post Keynesian model of credit rationing is to explain these developments in the context of fundamental uncertainty, and in particular to explain why a bank might generally increase interest rates for some borrowers, but refuse to raise them (and ration instead) for others. It is to that task that we now turn.

Some thoughts on a model

We assume that a bank is faced with requests for loans from a number of different borrowers. Denote the loan request from the i th borrower as d_i . The bank will form a subjective evaluation about the likelihood of repayment of the loan.¹² It will adhere to this evaluation with more or

¹¹ Jaffee and Modigliani (1969) also used the percentage of “safe” loans as a proxy for credit rationing.

¹² Note that the judgment about repayment is made about the borrower, not, as in New Keynesian theory, about individual projects. This enables the bank to explicitly consider questions of financial fragility that affect the borrower’s overall ability to repay, even though the success of a particular project may be unaffected.

less confidence—in Keynes's terms there will be a state of confidence associated with its expectation of the future return. Of course, both the estimate of the expected return and the confidence in this forecast will be the bank's best guesses about an uncertain future. They will be based on past history, on conventions, and on the mood of bankers at the time. They can be expected to change dramatically.

In contrast to New Keynesian theory, these guesses are not the bank's best estimate of a "true" probability distribution, a distribution that is known by the borrower but not by the bank. The "true" distribution is not known by the borrower, and not known by the bank, because as Crotty (1994) has expressed it, there is no "true" distribution "out there."¹³

The bank, though, has to make a decision on the loan request. We assume that in so doing it formulates an estimate of the borrower's cash flow that is available for debt service. The bank will have a state of confidence about this estimate. If the bank is highly confident in its estimate of the repayment prospects of a particular borrower, it will consider a loan to that borrower to be of low risk; on the other hand, if its state of confidence is low, the bank will consider the loan to be relatively risky.

The bank's state of confidence will enter into its decision-making process in two ways. First, riskier borrowers in whom the bank has relatively low confidence will be charged higher interest rates. Second, we assume that there is a minimum state of confidence that a bank must have in order to make a loan; below some limit, the bank will not make the loan.

There are two reasons for this. For one thing, it may be that there is some absolute limit to the amount of risk that the bank is willing to tolerate, despite the promised compensating return. Second, the bank is aware that the promised return is just that—a promise. If a bank reaches too far for a promised return, it could aggravate the condition of a financially fragile borrower. The result could be a realized return far short of what the bank desires—that is, default. As noted above, this is

¹³ For this reason, it seems that "probability distributions are not the basis for comprehending real world behavior under uncertainty" (Davidson, 1991, p. 130), and that it is not productive to try to incorporate ideas about fragility or confidence into New Keynesian probability distributions. For other perspectives on the question of complementarity between New Keynesian and Post Keynesian assumptions, see Van Ees and Garretsen (1993), Dymski (1993), Fazzari and Variato (1994), and Crotty (1996).

a situation where the price of a product can affect its quality: increasing the interest rate can erode the ability to repay debt.

So we assume that the bank uses the following decision rule: accept the loan request if

$$(1) \quad [CF_i / ((s_i + c) * L_i)] > R, \text{ subject to } SC_i > K;$$

otherwise reject (ration) the request. Here CF_i is the bank's best guess concerning the borrower's cash flow available for debt service; c is the bank's cost of funds; s_i is the spread over the bank's cost of funds (so that $s_i + c$ is the interest rate charged the borrower); L_i is the size of the loan; R is the minimum acceptable ratio of cash flow to interest payments; SC_i represents the bank's state of confidence in its estimate of CF_i ; and K , a measure of the bank's risk preference, is the limit below which the bank's state of confidence cannot fall if it is to make the loan.

For a particular borrower, the spread over the bank's cost of funds, s_i , varies inversely with SC_i . But as the bank becomes more risk-averse (K increases), the spread for all borrowers will increase, so that

$$(2) \quad s_i = s^{\text{MIN}} + b(K) / SC_i, \quad db/dK > 0.$$

Here s^{MIN} , b , K , and SC_i are taken to be positive, and s^{MIN} is the "risk-free" spread.

In the short run, s_i will not vary for borrowers with loan commitments, but over the longer term, as the terms of commitments are renegotiated, equation (2) will apply to these borrowers as well. In addition, we know that the bank's non-price terms also vary with the bank's judgment of borrower risk (SC_i). One particular nonprice term is of special interest in the credit rationing context, the size of credit lines. As the bank's state of confidence in a borrower decreases, we would expect that the maximum loan amount that the borrower could access, L_i^{MAX} , would tend to decrease.

K and R represent the bank's risk preferences and their judgments concerning prudent levels of debt commitments; they can be expected to change, sometimes dramatically, over time. Both K and R are directly reflected in the bank's minimum standards for extending a loan, so that borrowers who fail the test in equation (1) will also fail to meet these standards.

Credit rationing here refers to any situation in which the bank refuses to lend to a particular borrower, despite the borrower's willingness to pay a higher interest rate. (Note that this is a much broader conception

of credit rationing than the “pure” credit rationing of New Keynesian theory.) Credit rationing as here defined is due to the bank’s best guess about the probability of repayment. It is a result either of the bank’s view that the borrower’s cash flows will be insufficient to pay back the loan, or that the loan is too risky, in the sense that the bank has a low state of confidence in its ability to accurately predict what those cash flows might be.

It is now fairly straightforward to answer the questions posed at the end of the last section. If a bank’s state of confidence in a particular borrower or class of borrowers falls, it will increase both the price and nonprice terms of lending. Interest-rate spreads will increase. Some borrowers will qualify for these higher rates with no difficulties; some, though, may be able to borrow at the higher rate, but still be subject to some credit rationing by not being able to borrow all they want. But if, for example, the increased spread pushes the potential interest payments, $(s_i + c) * L_i$, too close to the bank’s estimate of likely cash flows (CF_i), the bank will refuse to lend to the borrower at this higher interest rate, despite the willingness of the borrower to pay the higher rate. The borrower will be rationed: the borrower will fail to meet minimum standards such as minimum interest coverage (cash flow to interest payment) ratios; indeed, this is in part why the bank judged the borrower’s expected cash flows inadequate in the first place.

Or it may be that the bank becomes more wary regarding its attitude toward risk (K increases). In this case, spreads on loans increase and the minimum standards for qualifying for a loan also increase. Again, some borrowers may be able to borrow at the higher spread, while others are rationed. In both of these examples interest-rate spreads, nonprice standards, and credit rationing increase together. If at the same time the Federal Reserve is tightening monetary policy, the bank’s cost of funds will also increase, and the general level of interest rates charged borrowers will also increase.

Clearly the borrower may, and probably will, have a different estimate of the relevant cash flows, and the state of confidence attached to that estimate. Credit rationing is due to these asymmetric *expectations* (and, in addition to financial fragility, as noted above). It is not due to asymmetric information in the New Keynesian sense, where the borrower knows the true probability density function while the bank doesn’t, but it is certainly compatible with the general phenomenon of asymmetric information.¹⁴ If

¹⁴ For an interpretation that incorporates adverse selection and credit rationing into a Post Keynesian framework, see Grabel (1995).

the borrower has more information than the bank about the possibility of voluntary default, as Keynes suggested, or if the bank suspects that higher interest rates may lead borrowers to take on greater risk after the loan is made, as in the New Keynesian scenario, the bank may want to increase R in order to compensate. We would expect R (and K) to change over the course of the business cycle and to be affected by such things as the bank's capital position, its history of loan losses, and its general level of profitability.

The bank's estimate of the expected cash flows, CF_t , as well as its state of confidence, SC_t , in those cash flows, will be influenced by conventional wisdom at any given time concerning the profitability of certain types of loans, such as those financing commercial real estate, the energy industry, and takeovers and leveraged buyouts. These variables, and also K and R , will be influenced as well by a general feeling of optimism or pessimism concerning the future, what Keynes called the "degree of optimism." As Minsky has pointed out, the general attitudes of bankers along these lines are likely to change systematically over the course of the business-cycle expansion, changing from highly optimistic in the boom stage of the expansion to more pessimistic as the expansion comes to an end.

In addition, the overall financial condition of borrowers, clearly central to the bank's judgment of the ability of borrowers to repay their loans, also deteriorates as the business-cycle expansion approaches its peak (Wolfson, 1994, pp. 153–159).

Some discussion is also in order concerning the cost of funds variable c . In the analysis above, we take c as given. This is a simplification, in three respects. First, we know that banks use several cost-of-funds interest rates, depending on the maturity and other characteristics of the loan. Second, we know that the markup on small loans is based on the prime rate, which is a bank-administered rate that is not per se a cost-of-funds variable—although it is true that increases in the discount rate have often been used by the banks as an opportunity to raise the prime rate (Epstein, 1990).

Third, the cost of funds will change over time, in particular as monetary policy changes or as banks use liability management. However, the cost-of-funds variable is not different from the other variables in this respect. We would expect all of them to change endogenously over time, in response to changing conditions and changing expectations. In this sense, the above analysis is more a framework with which to understand credit rationing than a definitive formal solution.

The notional demand curve

The above discussion implies that, given the borrower's demand for loans, the bank will first determine the amount of loan requests that it deems creditworthy. Thus, it will take the borrower's demand for credit to be a notional one; it will determine, from its perspective, the effective demand curve of creditworthy borrowers.

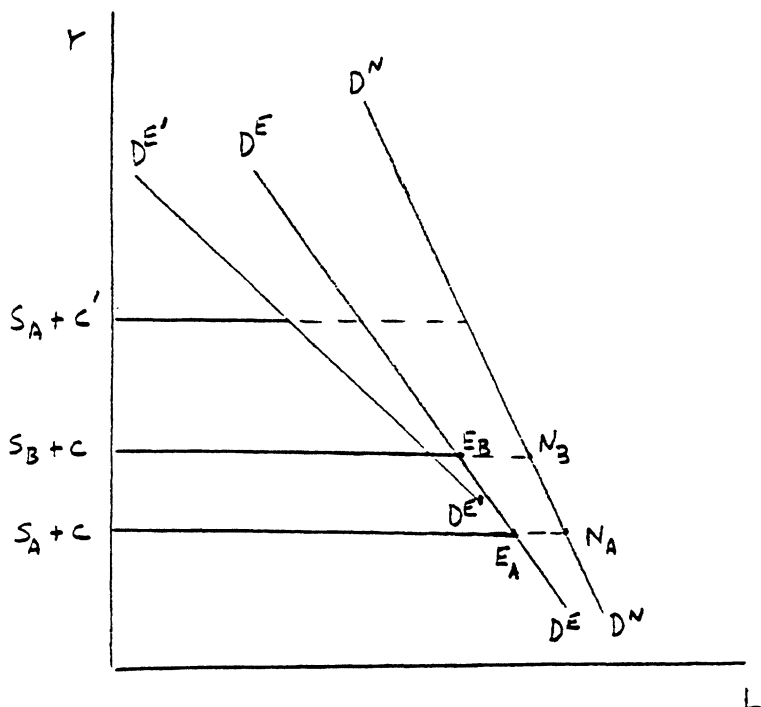
The difference between the notional and effective demand curves arises because the demand for loans is different from most other demands for goods and services in the economy. To purchase an ordinary commodity, all that is necessary is an amount of money equal to the price of the commodity. To purchase a loan, however, it is necessary to obtain the consent of the lender.

Given the effective demand curve, then, we assume that the bank does its best to accommodate all demands for loans. Here, as in the endogenous money case in general, the supply of loans is not independent of the demand. With credit rationing, the supply of loans depends not only on the borrower's demand for credit (the notional demand curve), but also on the proportion of that demand the bank judges to be creditworthy (the effective demand curve). If the bank has a greater supply of funds than necessary to meet effective loan demands, it will usually buy investment securities. If its supply of funds is insufficient, it will either sell securities or use liability management to borrow in the market for purchased funds (federal funds, Eurodollars, commercial paper, etc.).

The situation is depicted in figure 3. $D^N D^N$ is the notional demand curve faced by the bank (aggregated over all borrowers). We assume that $s_A + c$ is the average interest rate charged borrowers, and that at this interest rate the bank is willing to make loans represented by the point E_A ; $E_A N_A$ represents the amount of credit rationing. If the bank raises its average spread to s_B , then the amount of loans extended will be given by the point E_B , and credit rationing will increase to $E_B N_B$. The effective demand curve will be $D^E D^E$. As noted above, we are assuming that interest-rate spreads, nonprice terms, and credit rationing are all positively related.

The two curves are drawn for a given cost of funds, c . If the bank's cost of funds increases, the effective demand curve will shift up and to the left, to $D^{E'} D^{E'}$. At any particular spread, such as s_A , credit rationing would be expected to increase, since the higher level of interest rates would make debt repayment more difficult.

Figure 3 Credit rationing with asymmetric expectations: the notional demand curve and "endogenous money" supply curve



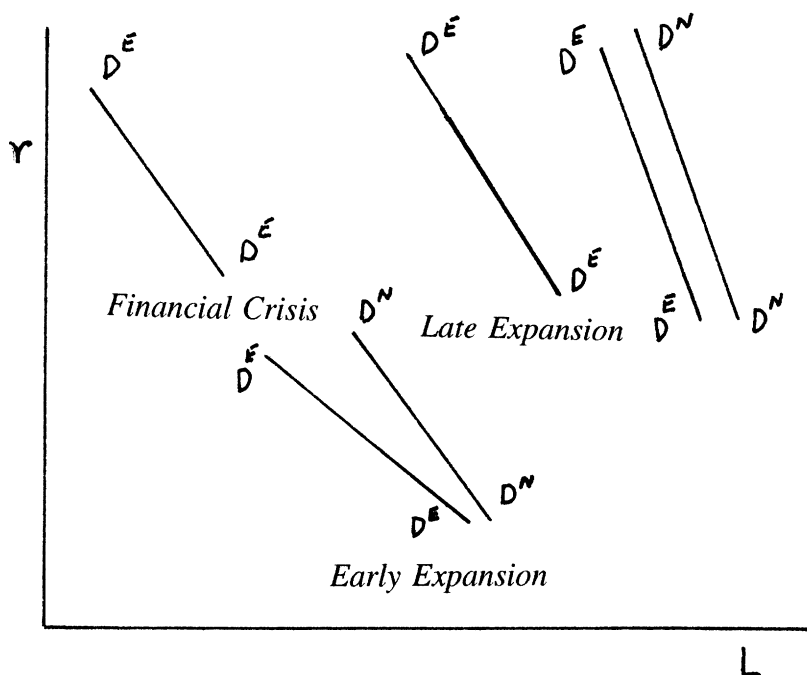
The role of expectations and the business cycle

However, the main "action" that we would expect in this model does not have to do with relatively small changes in interest rates at a given point in time. It has to do with changes in the factors that affect banks' expectations of the future. It is useful to look at three examples.

The first concerns the boom and bust in several areas of bank lending in the recent past. In the 1970s, banks were eager to lend to developing countries; in the 1980s, especially after Mexico's difficulties in 1982, banks were mainly trying to reduce their exposure. In the mid-1980s, banks lent heavily to finance hostile takeovers and commercial real estate; by the early 1990s, banks were retreating from these areas.

A second example is the quick collapse of lending that accompanies a financial crisis. A typical scenario follows from the shock to the financial system from a surprise failure or default on its obligations of some

Figure 4 Credit rationing and the business cycle



large financial or nonfinancial institution. Suddenly, all the variables in our model change, especially the state of confidence of expected returns (SC_i), the bank's tolerance for risk (K), and its judgment of prudent debt commitments (R). A sharp increase in credit rationing follows, which in part defines the nature of the financial crisis.

The third example is the typical pattern that occurs over the course of the business-cycle expansion. Typically, business profits decline and interest rates increase. As the peak of the expansion approaches and financial fragility worsens, bankers tend to revise their expectations of the future and of the desirable debt load that should be carried by borrowers. Credit rationing increases.

This situation is depicted in figure 4. In the early expansion, interest rates and the demand for loans are low. As the boom progresses, interest rates increase, but expectations typically become more euphoric, optimism increases, and the demand for loans shifts to the right. The spur to credit rationing due to higher interest rates is overwhelmed a higher estimation of cash flows (CF_i), a higher state of confidence (SC_i), and positive changes in risk preferences (K) and acceptable debt levels (R). Minimum standards fall, and the effective demand curve shifts to the

right and rotates clockwise. However, typically in the late expansion, as profits fall off relative to debt payments, bankers' attitudes become more pessimistic. An unexpected event can drastically change expectations and lead to the collapse of the supply of credit so characteristic of financial crises.

REFERENCES

- Bernanke, Ben S. "Credit in the Macroeconomy." *Federal Reserve Bank of New York Quarterly Review*, Spring 1993, 18, 50–70.
- Bernanke, Ben S., and Lown, Cara. "The Credit Crunch." *Brookings Papers on Economic Activity* 1992, 2, 205–239.
- Board of Governors of the Federal Reserve System. "Senior Loan Officer Opinion Survey on Bank Lending Practices." Prepared by William B. English with research assistance by Julius C. Clark. November 21, 1994.
- Brady, Thomas F. "Changes in Loan Pricing and Business Lending at Commercial Banks." *Federal Reserve Bulletin*, January 1985, 71, 1–13.
- Crotty, James. "Are Keynesian Uncertainty and Macrotheory Compatible? Conventional Decision Making, Institutional Structures, and Conditional Stability in Keynesian Macromodels." In G. Dymski and R. Pollin (eds.), *New Perspectives in Monetary Macroeconomics*. Ann Arbor: University of Michigan Press, 1994.
- . "Is New Keynesian Investment Theory Really 'Keynesian'? Reflections on Fazzari and Variato." *Journal of Post Keynesian Economics*, Spring 1996, 18 (3), this issue, 333–357.
- Davidson, Paul. "Is Probability Theory Relevant for Uncertainty? A Post Keynesian Perspective." *Journal of Economic Perspectives*, Winter 1991, 5, 129–144.
- Dymski, Gary. "A 'New View' of the Role of Banking Firms in Keynesian Monetary Theory." *Journal of Post Keynesian Economics*, Spring 1992, 14 (3), 311–320.
- . "Keynesian Uncertainty and Asymmetric Information: Complementary or Contradictory?" *Journal of Post Keynesian Economics*, Fall 1993, 16 (1), 49–54.
- Epstein, Gerald. "Prime Rates, Federal Reserve Signaling, and Financial Instability." *Journal of Post Keynesian Economics*, Summer 1990, 12 (4), 618–635.
- Fazzari, Steven, and Variato, Anna Maria. "Asymmetric Information and Keynesian Theories of Investment." *Journal of Post Keynesian Economics*, Spring 1994, 16 (3), 351–370.
- Friedman, Benjamin M., and Kuttner, Kenneth. "Indicator Properties of the Paper-Bill Spread: A Closer Look at the 1990–91 Recession." Mimeo, September 1994.
- Grabel, Ilene. "Speculation-led Economic Development: A Post Keynesian Interpretation of Financial Liberalization Programs in the Third World." *International Review of Applied Economics*, 1995, 9 (2), 127–149.
- Hamdani, Kausar; Rodrigues, Anthony P.; and Varvatsoulis, Maria. "Survey Evidence on Credit Tightening and the Factors Behind the Recent Credit Crunch." *Studies on Causes and Consequences of the 1989–92 Credit Slowdown*. Federal Reserve Bank of New York, 1994.
- Hansell, Saul. "Banks at the Dawn of New Worries." *The New York Times*, February 21, 1995, C1, C9.
- Jaffee, Dwight M., and Modigliani, Franco. "A Theory and Test of Credit Ration-

ing." *American Economic Review*, December 1969, 59, 850–872.

Jaffee, Dwight, and Stiglitz, Joseph. "Credit Rationing." In Benjamin M. Friedman and Frank H. Hahn (eds.), *Handbook of Monetary Economics*, vol. 2. New York: Elsevier Science Publishers, 1990, pp. 837–888.

Keynes, John Maynard. *A Treatise on Money*, 2 vols. New York: Harcourt, Brace, 1930.

Lang, William W., and Nakamura, Leonard I. "'Flight to Quality' in Bank Lending and Economic Activity." Federal Reserve Bank of Philadelphia Working Paper no. 92–20. October 1992.

Minsky, Hyman P. "The Financial Instability Hypothesis: An Interpretation of Keynes and an Alternative to 'Standard' Theory." *Challenge*, March–April 1977a, 20, 20–27.

———. "A Theory of Systemic Fragility." In Edward I. Altman and Arnold W. Sametz (eds.), *Financial Crises: Institutions and Markets in a Fragile Environment*. New York: John Wiley and Sons, 1977b, pp. 138–152.

———. "Finance and Profits: The Changing Nature of American Business Cycles." In U.S. Congress, Joint Economic Committee, *The Business Cycle and Public Policy, 1929–80*. Washington, DC: Government Printing Office, 1980, pp. 209–244.

———. *Stabilizing an Unstable Economy*. New Haven, CT: Yale University Press, 1986.

Stiglitz, Joseph. "The Causes and Consequences of the Dependence of Quality on Price." *Journal of Economic Literature*, March 1987, 25, 1–48.

Stiglitz, Joseph, and Weiss, Andrew. "Credit Rationing in Markets with Imperfect Information." *American Economic Review*, June 1981, 71, 393–410.

Tobin, James. *Asset Accumulation and Economic Activity*. Chicago: University of Chicago Press, 1980.

Van Ees, Hans, and Garretsen, Harry. "Financial Markets and the Complementarity of Asymmetric Information and Fundamental Uncertainty." *Journal of Post Keynesian Economics*, Fall 1993, 16 (1), 37–48.

Wolfson, Martin H. *Financial Crises: Understanding the Postwar U.S. Experience*, 2d ed. Armonk, NY: M.E. Sharpe, 1994.